

Learned closures and nonlinear manifolds for parametric projection-based reduced models

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Abstract

We compare learned closures and nonlinear manifolds for parametric projection-based reduced-order models of a nonlinear time-dependent system. All models use a common Euclidean proper-orthogonal-decomposition (POD) coordinate system and are trained from full-residual linear projection-based reduced-order model (PROM) trajectories. The comparison includes state-conditioned, parameter-time-conditioned, and hybrid PROM-ANN closures; an intrusive PROM-POD-AE decoder manifold; a non-intrusive POD-NN-ROM direct coefficient map; and a non-intrusive latent POD-DL-ROM.

The numerical study uses a parametric two-dimensional inviscid Burgers benchmark. The intrusive models all minimize the same time-discrete residual, but over different learned trial maps. A shared parameter-time map gives a controlled sweep from a non-intrusive POD-NN-ROM endpoint to the full-coordinate linear PROM. A complementary tail-perturbation diagnostic quantifies how a prescribed Case-2 secondary-coordinate error transfers to the residual-solved primary coordinates. We also state the empirical cubature method (ECM) formulation required for a future HPROM extension; all numerical results reported here are full-residual PROM results.

1 Introduction

Projection-based model order reduction (PMOR) reduces a high-dimensional model while retaining its governing residual and its parameter dependence [1, 2]. For nonlinear, transport-dominated systems, an accurate linear subspace may nevertheless require many coordinates. Nonlinear trial manifolds, learned closures, and latent representations seek to reduce the number of online unknowns without discarding the projection principle [3, 4, 5].

This work compares intrusive closure-enhanced PROMs (PROM-ANN), an intrusive decoder manifold (PROM-POD-AE), a non-intrusive direct coefficient map (POD-NN-ROM), and a non-intrusive latent map (POD-DL-ROM). PROM means that the full time-discrete residual is minimized online. It is not synonymous with a generic ROM: the non-intrusive baselines do not evaluate a residual online. Likewise, the HPROM formulation below is a sampled-residual extension of PROM, not a second name for the same computation.

The reported campaign uses a standard Euclidean POD basis and a common full-residual linear PROM teacher. This permits a direct comparison between Case 2, which prescribes a learned secondary-coordinate tail but solves a primary coordinate block, and POD-NN-ROM, which predicts the full coordinate vector from the same parameter-time map. Hyperreduction is developed explicitly because an ECM rule must be tied to the trial manifold and its LSPG tangent; HPROM numerical results are deliberately deferred until the corresponding campaign is available.

2 Projection Framework

2.1 Time-discrete residual and Euclidean POD basis

Consider a parametric nonlinear semi-discrete initial-value problem

$$\mathbf{M}(\boldsymbol{\mu})\dot{\mathbf{u}}(t; \boldsymbol{\mu}) + \mathbf{f}(\mathbf{u}(t; \boldsymbol{\mu}); \boldsymbol{\mu}) - \mathbf{g}(t; \boldsymbol{\mu}) = \mathbf{0}, \quad \mathbf{u}(0; \boldsymbol{\mu}) = \mathbf{u}_0(\boldsymbol{\mu}), \quad (1)$$

with $\boldsymbol{\mu} \in \mathcal{D}$. For the selected implicit integrator, let $\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu})$ be the residual at step $m + 1$, so the HDM solves

$$\mathbf{r}^{m+1}(\mathbf{u}^{m+1}; \boldsymbol{\mu}) = \mathbf{0}. \quad (2)$$

Let $\mathbf{u}_{\text{ref}}$ be the snapshot mean and let $\mathbf{S}_c$ collect centered HDM snapshots. Its thin singular-value decomposition

$$\mathbf{S}_c = \mathbf{U}\boldsymbol{\Sigma}\mathbf{Y}^T \quad (3)$$

defines the Euclidean POD basis

$$\mathbf{V}_{\text{tot}} = \mathbf{U}(:, 1 : n_{\text{tot}}), \quad \mathbf{V}_{\text{tot}}^T \mathbf{V}_{\text{tot}} = \mathbf{I}. \quad (4)$$

The affine reconstruction is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N(t; \boldsymbol{\mu}), \quad \mathbf{q}_N \in \mathbb{R}^{n_{\text{tot}}}. \quad (5)$$

2.2 Linear PROM and LSPG reference

The full-residual linear PROM is the time-discrete least-squares Petrov–Galerkin (LSPG) problem

$$\mathbf{q}_N^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^{n_{\text{tot}}}} \|\mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (6)$$

At a Gauss–Newton iterate, define

$$\mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \quad \mathbf{T}_N = \mathbf{V}_{\text{tot}}, \quad \mathbf{W}_N = \mathbf{J} \mathbf{T}_N. \quad (7)$$

The increment solves

$$\mathbf{W}_N^T \mathbf{W}_N \Delta \mathbf{q}_N = -\mathbf{W}_N^T \mathbf{r}. \quad (8)$$

Equation (6) generates all training targets in the PROM campaign.

2.3 Linear HPROM and ECM formulation

This subsection fixes the hyperreduction formulation needed for the later HPROM campaign. Let $\mathcal{E} = \{e_i\}_{i=1}^{N_e}$ be the mesh entities and assemble the residual from local contributions,

$$\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) = \sum_{e_i \in \mathcal{E}} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i}^+ \mathbf{u}; \boldsymbol{\mu}). \quad (9)$$

Here $\mathbf{L}_{e_i}$ extracts residual degrees of freedom and $\mathbf{L}_{e_i}^+$ extracts the stencil required for evaluation. The projected LSPG residual and Gauss–Newton matrix are entity sums,

$$\begin{aligned} \mathbf{g}_N &= \mathbf{W}_N^T \mathbf{r} = \sum_{e_i \in \mathcal{E}} (\mathbf{L}_{e_i} \mathbf{W}_N)^T \mathbf{r}_{e_i}(\mathbf{L}_{e_i}^+ \boldsymbol{\phi}_N; \boldsymbol{\mu}), \\ \mathbf{J}_N &= \mathbf{W}_N^T \mathbf{J} \mathbf{T}_N = \sum_{e_i \in \mathcal{E}} (\mathbf{L}_{e_i} \mathbf{W}_N)^T \mathbf{J}_{e_i}(\mathbf{L}_{e_i}^+ \mathbf{T}_N), \end{aligned} \quad (10)$$

where $\boldsymbol{\phi}_N(\mathbf{q}_N) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N$. ECM selects a reduced set $\tilde{\mathcal{E}}_N$ and positive weights ξ_{N, e_i} that replace both sums by weighted sums over $\tilde{\mathcal{E}}_N$.

For a selected parameter–time state s , the contribution of entity e_i is

$$\mathbf{c}_{N,e_i}^{(s)} = (\mathbf{L}_{e_i} \mathbf{W}_{N,s})^T \mathbf{r}_{e_i}(\mathbf{L}_{e_i}^+ \phi_N(\mathbf{q}_{N,s}); \boldsymbol{\mu}_s). \quad (11)$$

Stacking these vectors gives $\mathbf{C}_N$ and $\mathbf{b}_N = \mathbf{C}_N \mathbf{1}$. ECM seeks

$$\boldsymbol{\xi}_N = \arg \min_{\boldsymbol{\zeta} \geq 0} \|\mathbf{C}_N \boldsymbol{\zeta} - \mathbf{b}_N\|_2^2, \quad \tilde{\mathcal{E}}_N = \{e_i \in \mathcal{E} : \xi_{N,e_i} > 0\}. \quad (12)$$

We use ECM in the sense of empirical cubature [6, 7]. ECSW is another project–then–approximate hyperreduction strategy [8]; it is not the rule used here.

3 Learned Reduced Representations

3.1 PROM–ANN closures

Partition the common POD basis and coordinate vector as

$$\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \bar{\mathbf{V}}], \quad \mathbf{q}_N = \begin{bmatrix} \mathbf{q}_p \\ \mathbf{q}_s \end{bmatrix}, \quad n + \bar{n} = n_{\text{tot}}. \quad (13)$$

PROM–ANN supplies the secondary block through a learned relation

$$\mathbf{q}_s = \mathcal{F}(\mathbf{q}_p, \boldsymbol{\mu}, t), \quad \phi_{\mathcal{F}}(\mathbf{q}_p; \boldsymbol{\mu}, t) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}_p + \bar{\mathbf{V}} \mathcal{F}(\mathbf{q}_p, \boldsymbol{\mu}, t). \quad (14)$$

The network provides reduced secondary coordinates, not a direct HDM state approximation. The primary coordinates remain online unknowns of a residual minimization. This is the neural-network-augmented projection construction of [3]; analogous closure roles have also been represented by non-neural regressors [9].

The three cases are information patterns in this common construction:

$$\text{Case 1: } \mathbf{q}_s = \mathcal{N}(\mathbf{q}_p), \quad \text{Case 2: } \mathbf{q}_s = \mathcal{M}(\boldsymbol{\mu}, t), \quad \text{Case 3: } \mathbf{q}_s = \mathcal{H}(\mathbf{q}_p, \boldsymbol{\mu}, t). \quad (15)$$

Case 1 is state-conditioned, Case 2 parameter–time-conditioned, and Case 3 combines both inputs. The LSPG problem is

$$\mathbf{q}_p^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{r}^{m+1}(\phi_{\mathcal{F}}(\mathbf{x}; \boldsymbol{\mu}, t^{m+1}); \boldsymbol{\mu})\|_2^2, \quad (16)$$

with tangent

$$\mathbf{T}_{\mathcal{F}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{F}}{\partial \mathbf{q}_p}, \quad \mathbf{W}_{\mathcal{F}} = \mathbf{J} \mathbf{T}_{\mathcal{F}}. \quad (17)$$

For Case 2, $\partial \mathcal{M} / \partial \mathbf{q}_p = \mathbf{0}$ and $\mathbf{T}_{\mathcal{M}} = \mathbf{V}$. Its tangent is therefore the cheapest, but its prescribed secondary block is not corrected through a state-dependent tangent.

3.2 PROM–POD–AE and intrusive trial maps

An autoencoder maps the complete linear coordinate vector through a latent variable,

$$\mathbf{z} = \mathcal{E}(\mathbf{q}_N), \quad \hat{\mathbf{q}}_N = \mathcal{D}(\mathbf{z}), \quad \mathbf{z} \in \mathbb{R}^{n_z}. \quad (18)$$

PROM–POD–AE uses the decoder as an intrusive nonlinear trial manifold,

$$\phi_{\text{AE}}(\mathbf{z}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{z}), \quad \mathbf{T}_{\text{AE}} = \mathbf{V}_{\text{tot}} \frac{\partial \mathcal{D}}{\partial \mathbf{z}}. \quad (19)$$

Its latent coordinate minimizes the same residual as (16), but over $\mathbf{z}$ rather than $\mathbf{q}_p$. This is distinct from POD–DL–ROM, which is non-intrusive.

Table 1: Trial decoders and tangent bases of the intrusive models. Every row minimizes the same time-discrete residual over its displayed online variable.

Model	Online variable	Trial decoder	Tangent
Linear PROM/HPROM	$\mathbf{q}_N$	$\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N$	$\mathbf{V}_{\text{tot}}$
PROM-ANN Case 1	$\mathbf{q}_p$	$\mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}_p + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q}_p)$	$\mathbf{V} + \bar{\mathbf{V}} \partial \mathcal{N} / \partial \mathbf{q}_p$
PROM-ANN Case 2	$\mathbf{q}_p$	$\mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}_p + \bar{\mathbf{V}} \mathcal{M}(\boldsymbol{\mu}, t)$	$\mathbf{V}$
PROM-ANN Case 3	$\mathbf{q}_p$	$\mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}_p + \bar{\mathbf{V}} \mathcal{H}(\mathbf{q}_p, \boldsymbol{\mu}, t)$	$\mathbf{V} + \bar{\mathbf{V}} \partial \mathcal{H} / \partial \mathbf{q}_p$
PROM-POD-AE	$\mathbf{z}$	$\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{z})$	$\mathbf{V}_{\text{tot}} \partial \mathcal{D} / \partial \mathbf{z}$

3.3 Model-specific ECM construction

The HPROM extension does not reuse the linear rule in (12). For any intrusive map ϕ_c and tangent $\mathbf{T}_c$, the entity contributions use $\mathbf{W}_c = \mathbf{J} \mathbf{T}_c$. A manifold coordinate is first fitted at every selected HDM state,

$$\mathbf{y}_{c,s} = \arg \min_{\mathbf{y}} \left\| \mathbf{u}_s^{m_s+1} - \phi_c(\mathbf{y}; \boldsymbol{\mu}_s, t^{m_s+1}) \right\|_2^2. \quad (20)$$

For Case 2 this is linear least squares after subtracting the prescribed tail,

$$\mathbf{q}_{p2,s} = \mathbf{V}^\dagger \left[\mathbf{u}_s^{m_s+1} - \mathbf{u}_{\text{ref}} - \bar{\mathbf{V}} \mathcal{M}(\boldsymbol{\mu}_s, t^{m_s+1}) \right]. \quad (21)$$

For Cases 1 and 3 and for the decoder manifold, it is nonlinear least squares, for example with the Gauss-Newton update

$$\mathbf{y}_{c,s}^{(k+1)} = \mathbf{y}_{c,s}^{(k)} - \left[\frac{\partial \phi_c}{\partial \mathbf{y}} \right]^\dagger \left[\phi_c(\mathbf{y}_{c,s}^{(k)}; \boldsymbol{\mu}_s, t^{m_s+1}) - \mathbf{u}_s^{m_s+1} \right]. \quad (22)$$

At the fitted state,

$$\mathbf{c}_{c,e_i}^{(s)} = (\mathbf{L}_{e_i} \mathbf{W}_{c,s})^T \mathbf{r}_{e_i}(\mathbf{L}_{e_i}^\dagger \phi_c(\mathbf{y}_{c,s}; \boldsymbol{\mu}_s, t^{m_s+1}); \boldsymbol{\mu}_s). \quad (23)$$

Stacking these vectors produces $\mathbf{C}_c$ and $\mathbf{b}_c = \mathbf{C}_c \mathbf{1}$ as in (12); a positive ECM rule is then computed for each map. The trained closure or decoder, its tangent, and the parameter-time sample set are therefore part of a rule. A linear-HPROM rule, or one built for a different nonlinear map, cannot be reused [6, 7, 10].

3.4 Non-intrusive reference models

POD-NN-ROM is a direct map

$$\mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{G}_q(\boldsymbol{\mu}, t), \quad (24)$$

and does not solve a residual online. POD-DL-ROM uses a learned latent map and decoder,

$$\mathbf{z}(t; \boldsymbol{\mu}) \approx \mathcal{G}_z(\boldsymbol{\mu}, t), \quad \mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{D}(\mathcal{G}_z(\boldsymbol{\mu}, t)), \quad (25)$$

following the POD-DL-ROM principle [11]. These are direct reduced-coordinate regressors, not PROM residual solves.

4 Benchmark and PROM Protocol

4.1 Burgers problem and discretization

We use the two-dimensional parametric inviscid Burgers benchmark of [3],

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x; \boldsymbol{\mu}), \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \quad (26)$$

where

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2} u_x^2 \\ \frac{1}{2} u_x u_y \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2} u_x u_y \\ \frac{1}{2} u_y^2 \end{bmatrix}, \quad \mathbf{S}(x; \boldsymbol{\mu}) = \begin{bmatrix} 0.02 \exp(\mu_2 x) \\ 0 \end{bmatrix}. \quad (27)$$

The domain is $[0, 100]^2$, $T_f = 25$, and $\mathcal{D} = [4.25, 5.50] \times [0.015, 0.03]$. The HDM uses a cell-centered finite-volume Godunov discretization on a 250×250 grid ($N = 125,000$) and 500 uniform time steps with $\Delta t = 0.05$.

4.2 Data, validation, and online points

Nine HDM trajectories on a 3×3 tensor grid define the centered snapshot matrix for the Euclidean POD. We retain $n_{\text{tot}} = 151$ modes, which capture 99.9900% of its energy. At these same nine parameters we solve the full-residual linear PROM. The resulting PROM-generated trajectories, rather than HDM snapshots orthogonally projected after the fact, are the learning targets for every model. This comparison-control choice aligns every learned model with the reduced optimality problem used as its teacher; it is not the conventional purely non-intrusive workflow, which typically uses projected HDM labels.

Two complete 151-coordinate linear-PROM trajectories are reserved before training,

$$\boldsymbol{\mu}^{\text{val},1} = (4.5625, 0.02625), \quad \boldsymbol{\mu}^{\text{val},2} = (5.1875, 0.01875). \quad (28)$$

Because each held-out parameter contains all time samples, this is a parameter-generalization test rather than a random time-snapshot split. The four online evaluations are

$$\begin{aligned} \boldsymbol{\mu}^{(v)} &= (4.875, 0.0225), \quad \boldsymbol{\mu}^{(1)} = (4.560, 0.0190), \\ \boldsymbol{\mu}^{(2)} &= (5.190, 0.0260), \quad \boldsymbol{\mu}^{(3)} = (4.000, 0.0330). \end{aligned} \quad (29)$$

The first three are in-domain and $\boldsymbol{\mu}^{(3)}$ is extrapolatory. The validation locations are neither training nor online evaluation points.

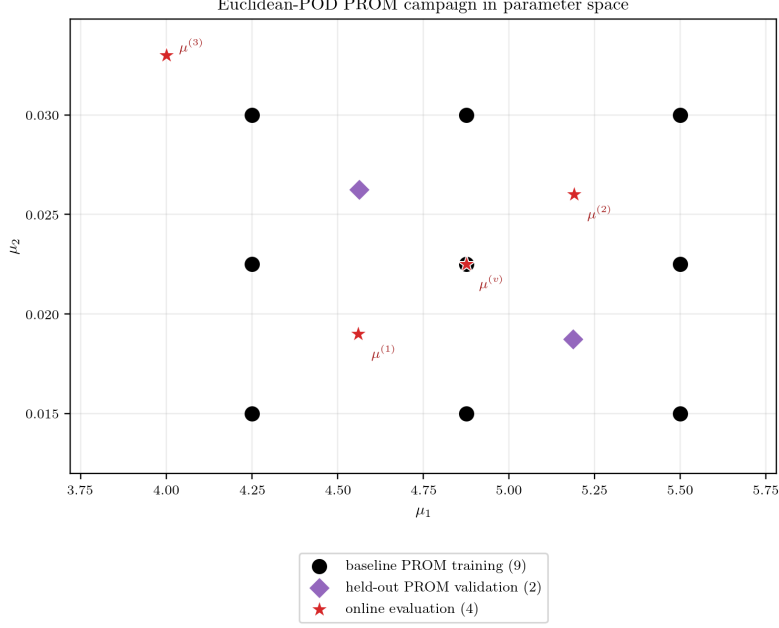


Figure 1: Euclidean-POD PROM data layout. Black circles are the nine HDM and linear-PROM training parameters; purple diamonds are the two held-out linear-PROM parameter-validation trajectories; red stars are the four online evaluation points. Validation targets are used only for regression model selection and are unrelated to the reported online tests.

4.3 Model training and shared Case-2 map

All networks use the same 4509 training and 1002 external-validation parameter-time samples. Case 1 and Case 3 use $n = 10$ primary coordinates; PROM-POD-AE and POD-DL-ROM use $n_z = 10$. The architectures and reduced coordinate errors are reported in Table 2.

Table 2: Training and held-out parameter-validation errors in reduced coordinates. The master map is shared between the Case-2 tail and the POD-NN-ROM direct endpoint.

Model	Learned map	Train e_q (%)	Validation e_q (%)	Parameters
PROM-ANN Case 1	$\mathbf{q}_p \mapsto \mathbf{q}_s$	1.422	3.590	564,621
Master map (Case 2 / POD-NN-ROM)	$(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_{151}$	2.888	4.261	565,399
PROM-ANN Case 3	$(\mathbf{q}_p, \boldsymbol{\mu}, t) \mapsto \mathbf{q}_s$	0.572	3.482	565,389
PROM-POD-AE	$\mathbf{q}_{151} \mapsto \mathbf{z} \mapsto \hat{\mathbf{q}}_{151}$	0.194	2.673	486,817
POD-DL-ROM	$(\boldsymbol{\mu}, t) \mapsto \mathbf{z} \mapsto \hat{\mathbf{q}}_{151}$	1.843	2.981	952,747

For the controlled Case-2 comparison, one master map is trained as

$$\mathcal{G}_q : (\boldsymbol{\mu}, t) \mapsto \hat{\mathbf{q}}_{151}. \quad (30)$$

For primary dimension n , Case 2 inserts $\mathbf{q}_s = \mathbf{P}_{>n} \mathcal{G}_q(\boldsymbol{\mu}, t)$ into the residual solve, whereas POD-NN-ROM uses the entire vector directly. Thus $n = 0$ is exactly the POD-NN-ROM endpoint from this same trained map, and $n = 151$ is the linear PROM endpoint. The shared-map construction is a diagnostic control, not a restriction of the general Case-2 formulation.

4.4 Reported errors

For each evaluation parameter, the state error is

$$e_u(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F}{\|\mathbf{u}_{\text{HDM}}\|_F}, \quad (31)$$

and the coefficient error against the full linear PROM trajectory is

$$e_q(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{q}_{151}^{\text{lin}} - \hat{\mathbf{q}}_{151}\|_F}{\|\mathbf{q}_{151}^{\text{lin}}\|_F}. \quad (32)$$

The linear PROM is an accuracy reference for the fixed basis and full residual, not a universal lower bound with respect to the HDM.

5 Euclidean-POD PROM Results

5.1 Baseline state accuracy and solution fields

Table 3 reports space–time state errors. The linear PROM remains below 0.5% at the three in-domain points. Among the learned intrusive models, Case 3 and PROM–POD–AE remain closest to that reference in domain. Case 2 improves on the direct POD–NN–ROM endpoint at all four points, but its prescribed tail remains visible, particularly away from the training grid. POD–DL–ROM is the closer of the two non-intrusive baselines in this campaign.

Table 3: Online state errors $100 \|\mathbf{u}_{\text{HDM}} - \mathbf{u}_{\text{ROM}}\|_F / \|\mathbf{u}_{\text{HDM}}\|_F$. The mean averages the three in-domain points. The horizontal rule separates intrusive residual-minimizing models from non-intrusive maps.

Model	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
Linear PROM	0.410	0.450	0.463	0.441	0.850
PROM–ANN Case 1	0.418	0.543	0.736	0.566	1.673
PROM–ANN Case 2 ($n = 10$)	1.163	1.900	1.758	1.607	2.696
PROM–ANN Case 3	0.408	0.512	0.563	0.494	1.521
PROM–POD–AE ($n_z = 10$)	0.411	0.533	0.711	0.551	1.754
POD–NN–ROM	1.187	1.918	1.989	1.698	4.351
POD–DL–ROM ($n_z = 10$)	0.827	1.178	1.370	1.125	2.267

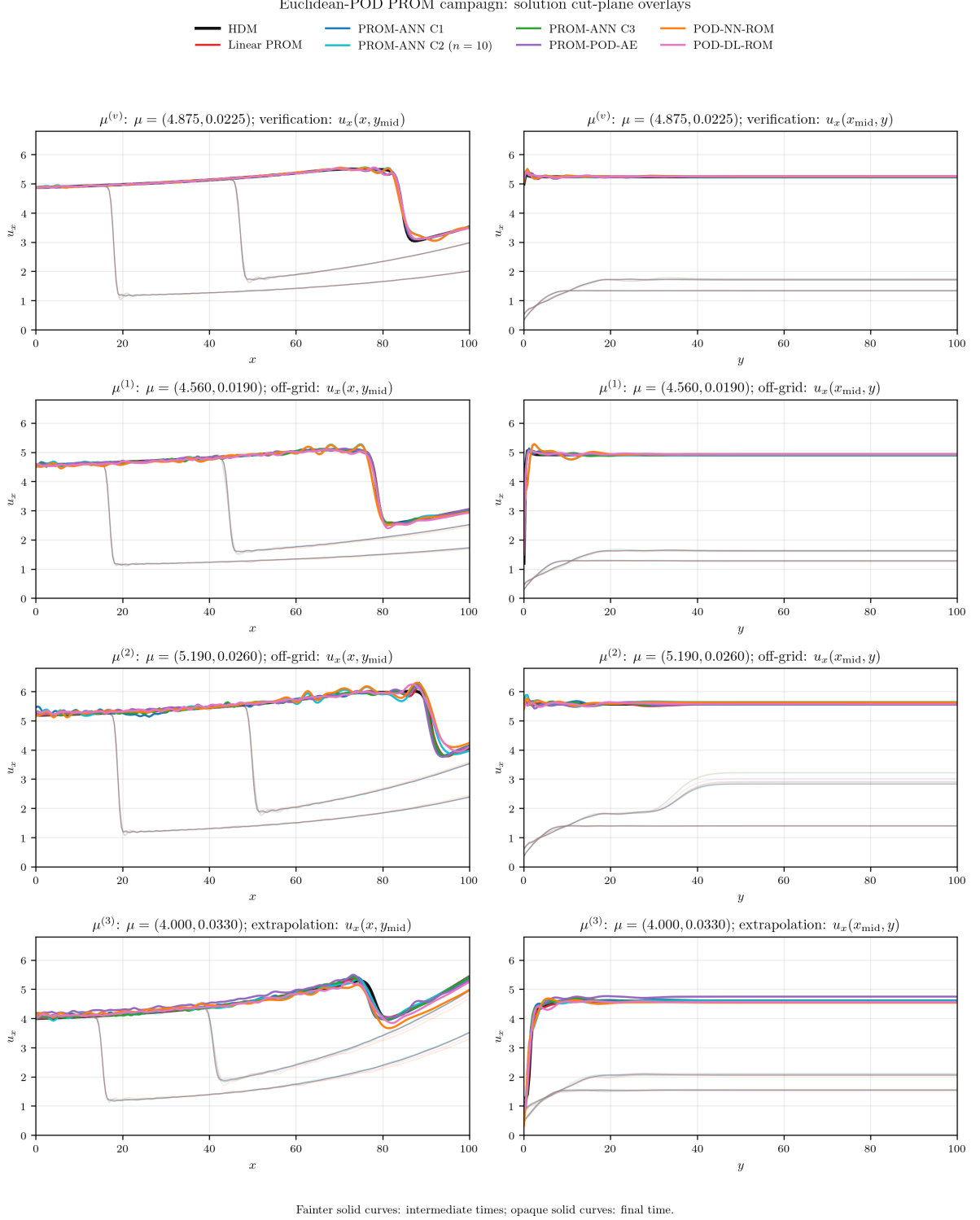


Figure 2: Final-time solution cut-plane overlays for the Euclidean-POD PROM campaign. Fainter solid curves show two intermediate times. Every intrusive curve uses the full residual; the two non-intrusive curves are direct reduced coordinate predictions.

5.2 Coefficient diagnostics

Table 4 and Fig. 3 compare each complete coordinate history with its 151-coordinate linear-PROM counterpart. The vertical line marks the $n = 10$ primary/secondary split used by

PROM-ANN. The direct maps are kept in the same figures and tables as the intrusive models, with an explicit separator, because all use the same linear-PROM coordinate teacher in this campaign.

Table 4: Relative coefficient-history errors e_q against the corresponding linear PROM. The horizontal rule separates intrusive and non-intrusive models, consistently with Table 3.

Model	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
Linear PROM	0.000	0.000	0.000	0.000	0.000
PROM-ANN Case 1	0.324	0.751	1.391	0.822	3.304
PROM-ANN Case 2 ($n = 10$)	2.653	4.346	3.956	3.652	6.234
PROM-ANN Case 3	0.222	0.710	0.798	0.577	2.644
PROM-POD-AE ($n_z = 10$)	0.316	0.804	1.290	0.803	3.664
POD-NN-ROM	2.713	4.378	4.494	3.861	10.214
POD-DL-ROM ($n_z = 10$)	1.811	2.611	3.031	2.484	5.127

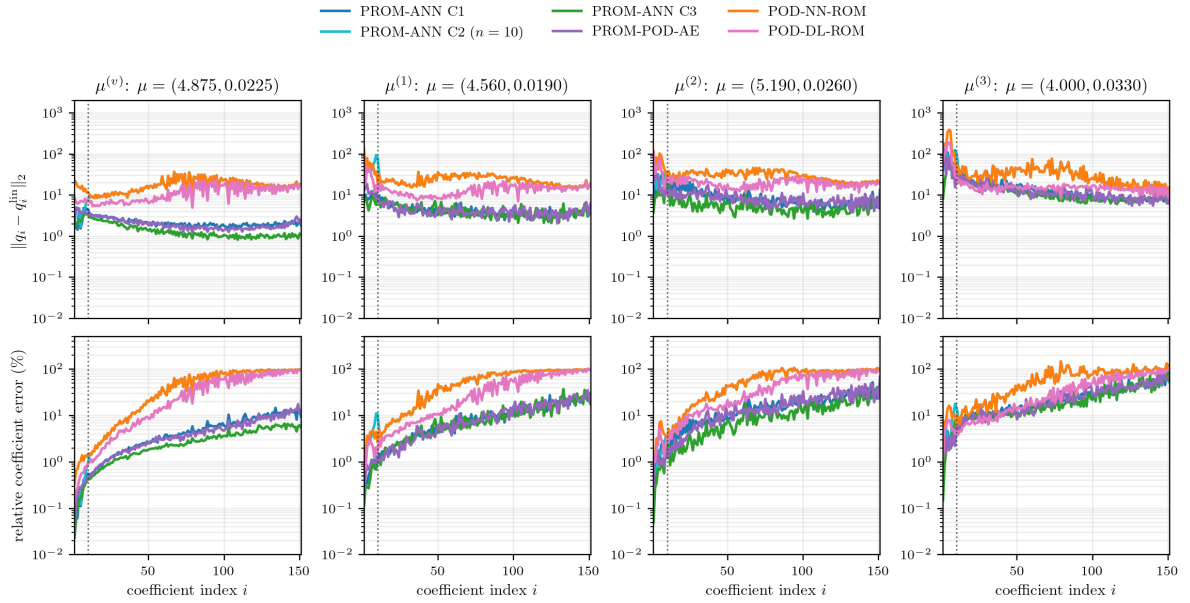


Figure 3: Per-coefficient absolute (top) and relative (bottom) errors against the 151-coordinate linear PROM. All panels in each row share the same logarithmic ordinate range. The dotted vertical line marks coefficient 10; all model curves are solid.

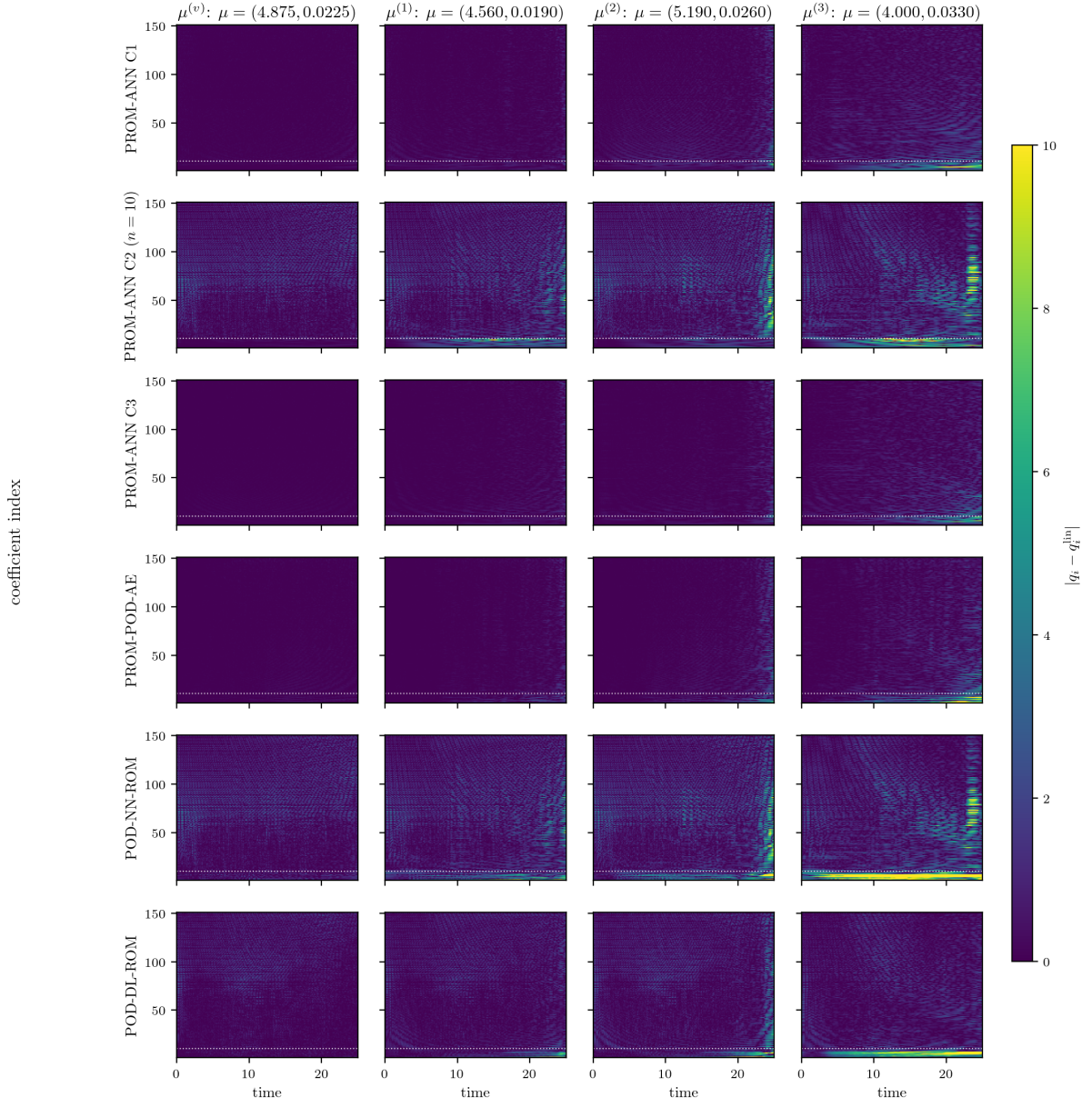


Figure 4: Absolute coefficient errors over time. Complete model names distinguish the intrusive PROM-POD-AE model from the non-intrusive POD-DL-ROM. The dotted white line marks the $n = 10$ primary split.

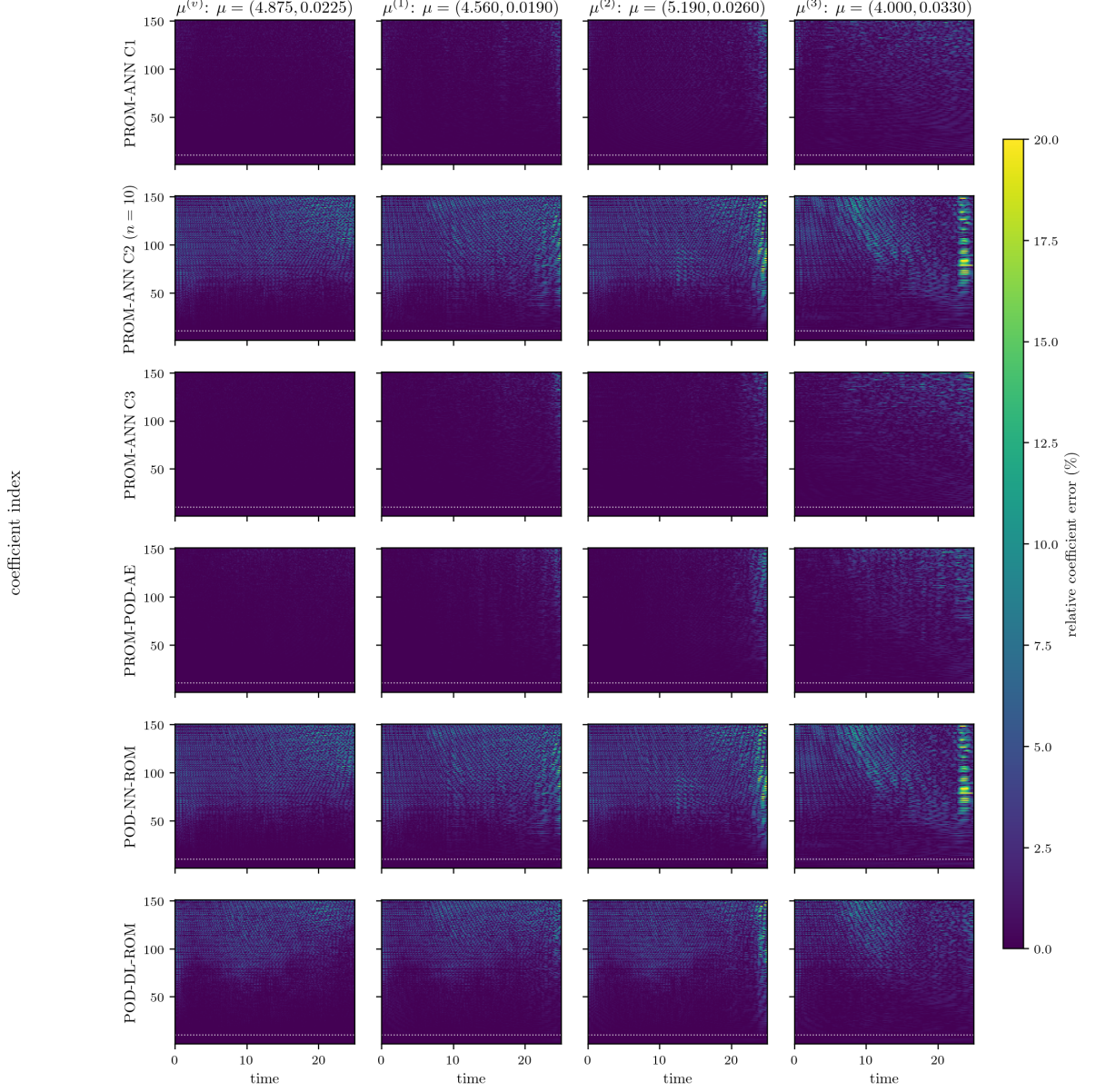


Figure 5: Relative coefficient errors over time, using the same ordering and coordinate split as Fig. 4.

5.3 Case-2 solved-dimension sweep

The shared master map permits an interpolation between a direct parameter-time regression and the full-coordinate linear PROM. Table 5 and Fig. 6 show that increasing n reduces state error, with the sharpest improvement between $n = 50$ and $n = 100$. This records the balance in this controlled Case-2 setting between coordinates supplied by the ANN and coordinates corrected by the residual.

Table 5: Case-2 state errors as the number of residual-solved primary coordinates is varied. The $n = 0$ and $n = 151$ endpoints are respectively the shared-map POD-NN-ROM and the linear PROM.

Solved n	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	Mean	$\mu^{(3)}$
0 (POD-NN-ROM)	1.187	1.918	1.989	1.698	4.351
10	1.163	1.900	1.758	1.607	2.696
20	1.158	1.556	1.665	1.460	2.201
30	1.154	1.529	1.601	1.428	2.178
50	1.161	1.506	1.486	1.384	2.036
100	0.785	0.958	0.863	0.869	1.183
151 (linear PROM)	0.410	0.450	0.463	0.441	0.850

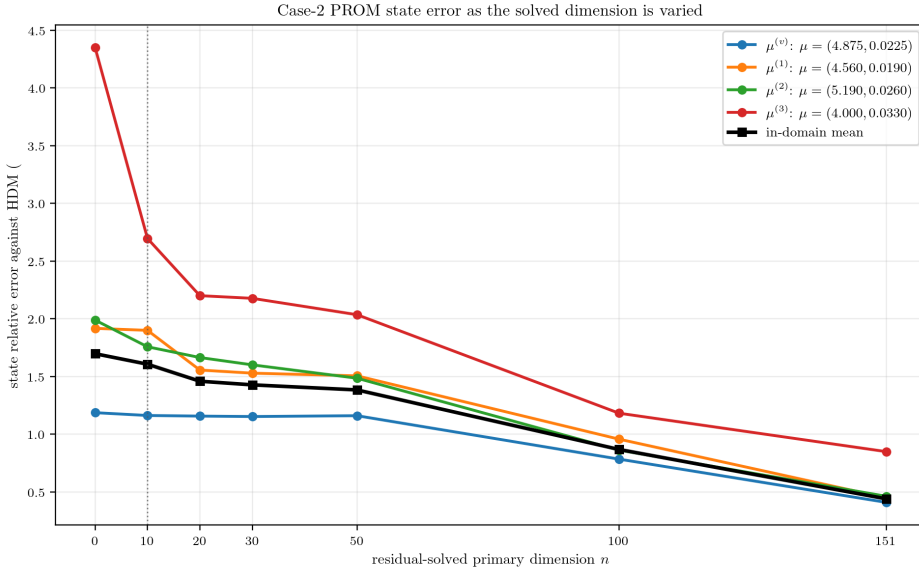


Figure 6: Case-2 state error versus the solved primary dimension. The black curve is the in-domain mean and the dotted line marks the baseline $n = 10$ split.

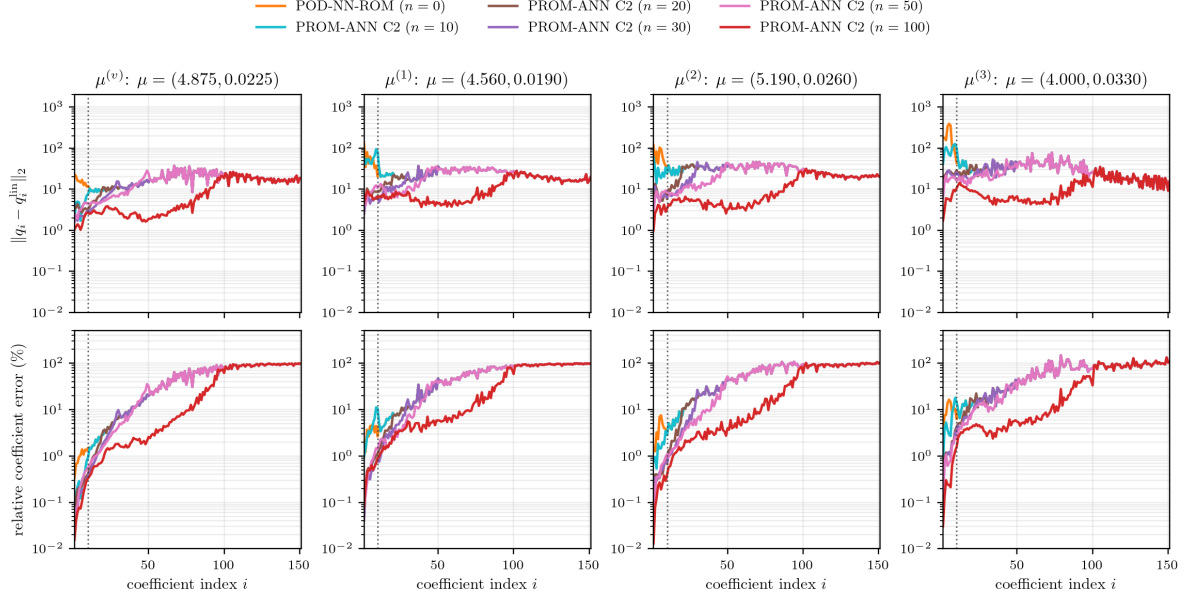


Figure 7: Case-2 coefficient trajectories in the shared-master-map sweep. Top: absolute error against linear PROM. Bottom: the corresponding relative error. The $n = 151$ linear-PROM reference has zero coefficient error and is therefore omitted.

5.4 Controlled Case-2 tail perturbation

To test the transfer mechanism independently of a regression magnitude, we start from the linear-PROM secondary history $\mathbf{q}_s^{\text{lin}}$ and add a perturbation in the direction of the actual master-map secondary error:

$$\mathbf{q}_{s\epsilon} = \mathbf{q}_s^{\text{lin}} + \frac{\epsilon}{100} \frac{\widehat{\mathbf{q}}_s - \mathbf{q}_s^{\text{lin}}}{\|\widehat{\mathbf{q}}_s - \mathbf{q}_s^{\text{lin}}\|_F} \|\mathbf{q}_s^{\text{lin}}\|_F. \quad (33)$$

For each prescribed percentage ϵ , Case 2 solves only the first ten coordinates. Thus the diagnostic uses the observed ANN-error direction but does not conflate its effect with a change of learned model. Table 6 and Fig. 8 show a monotone transfer from secondary-tail error to the recovered primary block. The amount of transfer is parameter dependent, so a tail loss alone does not imply a pointwise primary-coordinate bound.

Table 6: Primary-coordinate error induced by a controlled secondary-tail perturbation in Case 2 with $n = 10$. Each entry is relative to the corresponding linear-PROM primary history.

Imposed tail error (%)	$\mu^{(v)}$	$\mu^{(1)}$	$\mu^{(2)}$	$\mu^{(3)}$
0	0.001	0.001	0.001	0.000
1	0.131	0.095	0.060	0.098
3	0.382	0.283	0.169	0.291
5	0.617	0.469	0.265	0.480
10	1.133	0.922	0.453	0.940
15	1.544	1.364	0.602	1.389
20	1.856	1.801	0.773	1.839
30	2.281	2.702	1.413	2.778
50	4.034	4.949	4.208	5.044

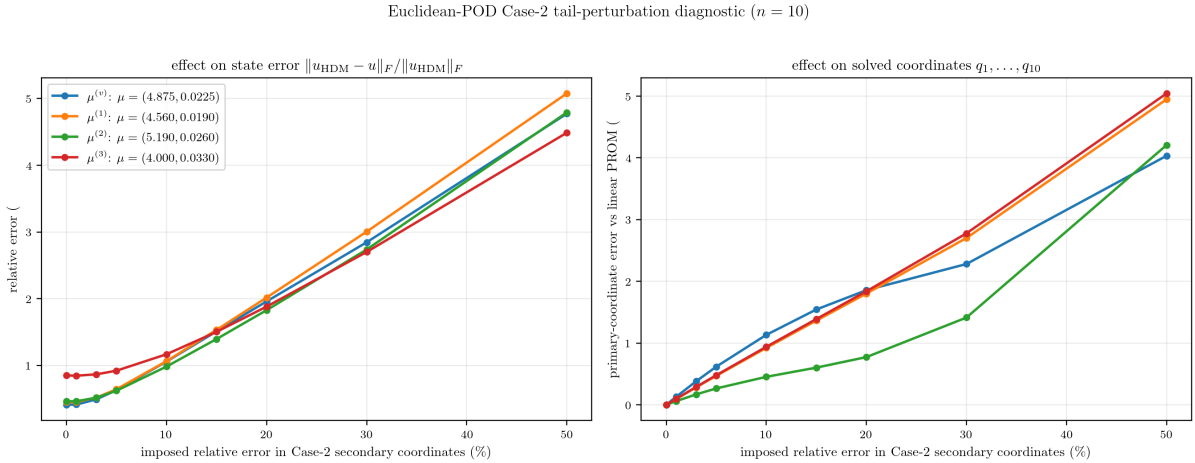


Figure 8: Euclidean-POD Case-2 sensitivity to a prescribed secondary-tail error, restricted to perturbations up to 50%. Left: state error against the HDM. Right: error in the ten residual-solved primary coordinates against linear PROM.

6 Conclusions

The Euclidean-POD PROM campaign gives a common full-residual comparison of three learned closures, an intrusive decoder, and two direct reduced-coordinate maps. The linear PROM remains the most accurate model in the fixed 151-coordinate trial representation. Case 3 and PROM-POD-AE retain close in-domain state accuracy while solving a low-dimensional online variable. The Case-2/POD-NN shared-map study gives a precise interpretation of intrusive correction: solving progressively more coordinates through the residual moves the result from the direct-map endpoint toward linear PROM, but the prescribed tail can still transfer its error into the solved block.

The results intentionally do not mix PROM evidence with an unavailable HPROM data

set. The ECM derivation establishes the required model-specific construction: an HPRM rule must be built from each learned trial map and its tangent, rather than borrowed from a linear HPRM or from a different closure. The next HPRM campaign can therefore use the same reporting order, tables, and model comparisons as the PROM study.

A LSPG-Sensitive Basis Exploration

Before adopting the Euclidean POD basis used in the main text, we explored an M -regularized basis metric intended to emphasize sensitivity of the LSPG normal equations to reduced-coordinate perturbations. The objective was to reduce transfer from a learned secondary block to residual-solved primary coordinates without degrading the linear reference. In the present Burgers campaign, controlled tests did not establish a systematic improvement sufficient to justify making that metric the main experimental variable. For clarity and reproducibility, every reported result therefore uses the standard Euclidean POD basis of (4). The metric-aware construction is retained as an exploratory direction, not as a contribution or as a source of the numerical comparisons in this paper.

References

- [1] Athanasios C Antoulas. *Approximation of large-scale dynamical systems*. SIAM, 2005.
- [2] Alfio Quarteroni, Gianluigi Rozza, et al. *Reduced order methods for modeling and computational reduction*, volume 9. Springer, 2014.
- [3] Joshua Barnett, Charbel Farhat, and Yvon Maday. Neural-network-augmented projection-based model order reduction for mitigating the Kolmogorov barrier to reducibility. *Journal of Computational Physics*, 492:112420, 2023.
- [4] Albert Cohen, Charbel Farhat, Yvon Maday, and Agustin Somacal. Nonlinear compressive reduced basis approximation for PDEs. *Comptes Rendus. Mécanique*, 351(S1):357–374, 2023.
- [5] Joubine Aghili, Hassan Ballout, Yvon Maday, and Christophe Prud’Homme. Nonlinear compressive reduced basis approximation: when taylor meets kolmogorov. *arXiv preprint arXiv:2601.13712*, 2026.
- [6] Joaquin Alberto Hernandez, Manuel Alejandro Caicedo, and Alex Ferrer. Dimensional hyper-reduction of nonlinear finite element models via empirical cubature. *Computer Methods in Applied Mechanics and Engineering*, 313:687–722, 2017.
- [7] JA Hernández. A multiscale method for periodic structures using domain decomposition and ecm-hyperreduction. *Computer Methods in Applied Mechanics and Engineering*, 368:113192, 2020.
- [8] Charbel Farhat, Todd Chapman, and Philip Avery. Structure-preserving, stability, and accuracy properties of the energy-conserving sampling and weighting method for the hyper reduction of nonlinear finite element dynamic models. *International Journal for Numerical Methods in Engineering*, 102(5):1077–1110, 2015.
- [9] Sebastian Ares De Parga, Radek Tezaur, Carlos G. Hernández, and Charbel Farhat. Nonlinear projection-based model order reduction with machine learning regression for closure error modeling in the latent space. *Computer Methods in Applied Mechanics and Engineering*, 448:118443, 2026.

- [10] S Ares De Parga, Jose Raul Bravo, JA Hernández, Rubén Zorrilla, and Riccardo Rossi. Hyper-reduction for petrov–galerkin reduced order models. *Computer Methods in Applied Mechanics and Engineering*, 416:116298, 2023.
- [11] Stefania Fresca and Andrea Manzoni. Pod-dl-rom: Enhancing deep learning-based reduced order models for nonlinear parametrized pdes by proper orthogonal decomposition. *Computer Methods in Applied Mechanics and Engineering*, 388:114181, 2022.